

Shell and Tube Heat Exchanger Model

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1 Overview

Shell and tube heat exchangers can be widely used in industry, in places such as oil refineries, power plants, and chemical processing facilities. Within a shell and tube heat exchanger, heat is transferred between two fluids that are separated and not mixed. One fluid flows inside the tubes, and the other flows through the surrounding shell at a different temperature. Through this model, it is possible to design a simple shell and tube heat exchanger and generate output results.

2 Methods and Equations

The fundamental thermodynamic behavior is governed by the Log Mean Temperature Difference (LMTD) method and the Effectiveness-NTU (ϵ -NTU) method.

2.1 Effectiveness-NTU Method

When given an existing heat exchanger design, the NTU (number of transfer units) method allows us to predict the outlet temperatures.

The value for NTU is defined as:

$$NTU = \frac{UA_s}{C_{\min}} \quad (1)$$

Where:

- U is the overall heat transfer coefficient ($\text{W}/(\text{m}^2 \cdot \text{K})$)
- A_s is the heat transfer surface area (m^2)

$$A_s = N \cdot \pi \cdot D_o \cdot L \quad (2)$$

- $C_{\min}$ is the minimum heat capacity rate (W/K), defined as the smaller value between the hot and cold fluid heat capacity rates ($\min(C_H, C_C)$)

The maximum heat transfer rate ($Q_{\max}$ in W) is defined as:

$$Q_{\max} = C_{\min}(T_{H,\text{in}} - T_{C,\text{in}}) \quad (3)$$

The effectiveness can be shown as:

$$\epsilon = \frac{Q_{\text{actual}}}{Q_{\max}} \quad (4)$$

The outlet temperatures of the heat exchanger can be found via the first law of thermodynamics:

$$Q_{\text{actual}} = \dot{m}_h \cdot C_{p,h} \cdot (T_{h,\text{in}} - T_{h,\text{out}}) \quad (5)$$

$$Q_{\text{actual}} = \dot{m}_c \cdot C_{p,c} \cdot (T_{c,\text{out}} - T_{c,\text{in}}) \quad (6)$$

Where:

- $\dot{m}$ is the mass flow rate (kg/s)
- C_p is the specific heat capacity ($\text{J}/(\text{kg} \cdot \text{K})$)

2.2 Log Mean Temperature Difference Method (LMTD)

When trying to determine the size of a heat exchanger to achieve a desired output temperature the LMTD method is used.

The total heat transfer rate Q (W) is defined as:

$$Q = U \cdot A_s \cdot F \cdot \Delta T_{lm} \quad (7)$$

Where:

- F is defined as the correction factor (Dimensionless).
- ΔT_{lm} (K) is defined as:

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)} \quad (8)$$

- For parallel flow: $\Delta T_1 = T_{h,in} - T_{c,in}$ and $\Delta T_2 = T_{h,out} - T_{c,out}$
- For counter flow: $\Delta T_1 = T_{h,in} - T_{c,out}$ and $\Delta T_2 = T_{h,out} - T_{c,in}$

The correction factor F is dependent on the geometry and temperatures of the heat exchanger.* The correction factor demonstrates the difference of the ΔT_{lm} from a counterflow arrangement*. For a single pass shell and tube heat exchanger we can assume that $F=1$. For other heat exchanger arrangements, in order to find the correction factor we must first calculate values P and R .

$$P = \frac{t_2 - t_1}{T_1 - t_1} \quad (9)$$

$$R = \frac{T_1 - T_2}{t_2 - t_1} \quad (10)$$

Once P and R have been calculated, reference correction factor charts to get an estimated F value.

3 Takeaways

It is important to note that shell and tube heat exchangers can become much more complex than what the model shows. Baffles, fins, and other implementations can be added to the heat exchangers to increase heat transfer. This model is intended to show the general behavior of a simple shell and tube heat exchanger

4 References and Data Sources

1. Bowman, R. A., Mueller, A. C., & Nagle, W. M. (1940). Mean Temperature Difference in Design. Transactions of the American Society of Mechanical Engineers, 62(4), 283–293. <https://doi.org/10.1115/1.4021450>
2. Çengel, Y. A., & Ghajar, A. J. (2015). Heat and mass transfer: Fundamentals & applications (5th ed.). McGraw-Hill Education.
3. The Engineering ToolBox (2009). Heat Exchanger Materials - Thermal Conductivities. [online] Available at: <https://www.engineeringtoolbox.com/heat-exchanger-material-thermal-conductivities-d1488.html>